Prediction Report

Bank Marketing Prediction using Logistic Regression

Objective

The objective of this project is to build a machine learning model that predicts whether a customer will subscribe to a bank term deposit based on demographic and financial information. The prediction helps the bank identify potential customers and improve the effectiveness of marketing campaigns.

Dataset

- **Dataset Name:** Bank Marketing Dataset
- **File:** bank.csv
- **Target Variable:** deposit
- **Prediction Type:** Binary Classification (Yes/No)

Data Preprocessing

The following preprocessing steps were performed:

- Loaded the dataset using Pandas.
- Checked for missing values.
- Separated features and target variable.
- Applied One-Hot Encoding to categorical features.
- Standardized numerical features using StandardScaler.
- Split the dataset into training (80%) and testing (20%) sets.

Model Used

A **Logistic Regression** model was selected because the target variable (`deposit`) contains two classes: **Yes** and **No**. Logistic Regression is a suitable algorithm for binary classification problems.

Model Training

The processed training data was used to train the Logistic Regression model. The trained model was then tested on unseen data to evaluate its predictive performance.

Evaluation Metrics

The following metrics were used:

- Accuracy Score
- Precision
- Recall
- F1-Score
- Confusion Matrix

These metrics provide a comprehensive evaluation of the model's classification performance.

Prediction

The trained model predicts whether a customer is likely to subscribe to a bank term deposit based on the provided customer information.

Model Saving

The trained model was saved using the Joblib library as:

bank_model.pkl

This saved model can be loaded later for making predictions without retraining.

Conclusion

The Logistic Regression model successfully predicts customer subscription behavior using the Bank Marketing dataset. Such predictive models can help banks target customers more efficiently, reduce marketing costs, and improve campaign success rates.